

Chapter 3 :- Mathematical Foundation for Machine Learning

- 1) Linear Algebra -
- 2) Statistics
- 3) Differential calculus

A) Linear Algebra

- Scalars, Vectors, Vector operation, Matrices, Matrix operation, Functions, Linear transformation, Inverse function, Eigen Values, Eigen Vectors

ex. Neural Networks

Definition:-

- Linear Algebra is branch of mathematics focus on study of vectors, vector space, linear transformation and system of linear equation.

Applications of Linear Algebra

a) Data Representation and manipulation.

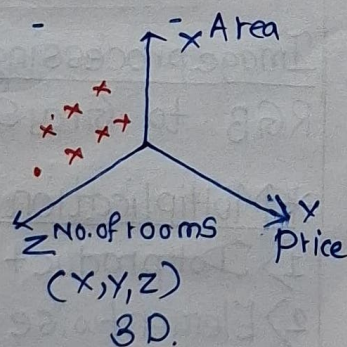
ex. House price prediction.

Area No. of Location Price
rooms

[1200 2 Nashik 45L]



$(x, y) = (1200, 45)$
2-D

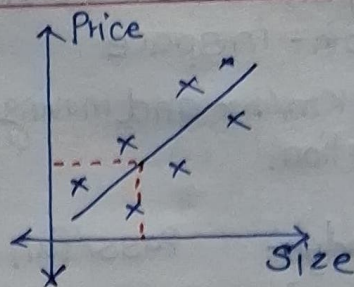


(x, y, z)
3-D.

b) Machine Learning and artificial Intelligence.

Model Train:- Matrix multiplication.

Matrix Arithmetic operations.



$$y = mx + c$$

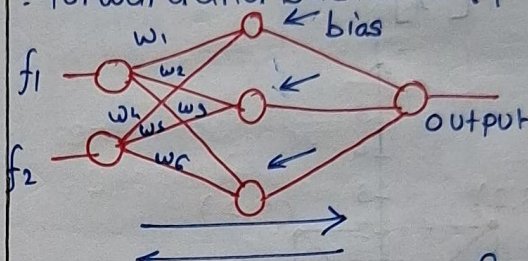
Dimensionality Reduction

:- PCA (Principal Component Analysis)

Reduce Higher Dimension $\rightarrow$ Lower Dimension

Neural Network

:- forward and Backward propagation



$f_1 \rightarrow$ Area $f_2 \rightarrow$ No. of rooms

Output :- price.

$$\begin{bmatrix} f_1 \\ f_2 \end{bmatrix} \begin{bmatrix} w_1 & w_2 & w_3 \\ w_4 & w_5 & w_6 \end{bmatrix} \Rightarrow \text{Matrix Multiplication}$$

Computer Graphics :- Image scaling (Linear Transformation)

Optimization:-

$$y = mx + c \leftarrow \text{intercept.}$$

Slope. for better prediction. we have to choose Right slope and intercept.

* Scalars and Vectors

Scalar is single numerical value, It represent magnitude

Applications in Data Science

- Simple linear Regressions

$$y = mx + c \rightarrow \text{intercept} \downarrow \text{Scalar value.}$$

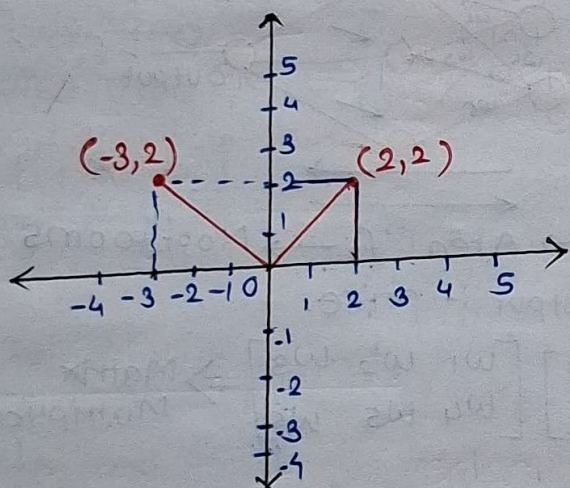
Vectors :- Magnitude + Direction.
 it represent point in space
 ex. car has 45 Km/hr and moving in east direction.

ex.

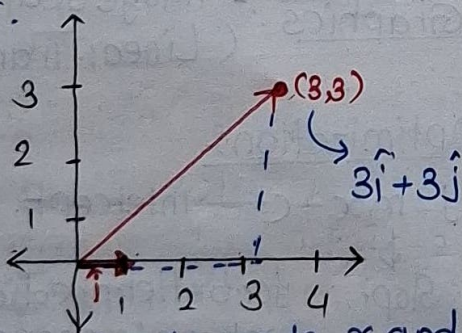
IQ	No. of student Hrs	Pass/Fail
[90	3hrs]	Fail
[100	3hrs]	Pass

Vector representing Person IQ and no. of Study Hrs = $[90, 3hrs]$

$$A = \begin{bmatrix} 90 \\ 3 \end{bmatrix} \quad B = \begin{bmatrix} 100 \\ 3 \end{bmatrix}$$



* Unit Vector.
 $(\hat{u} = 1)$

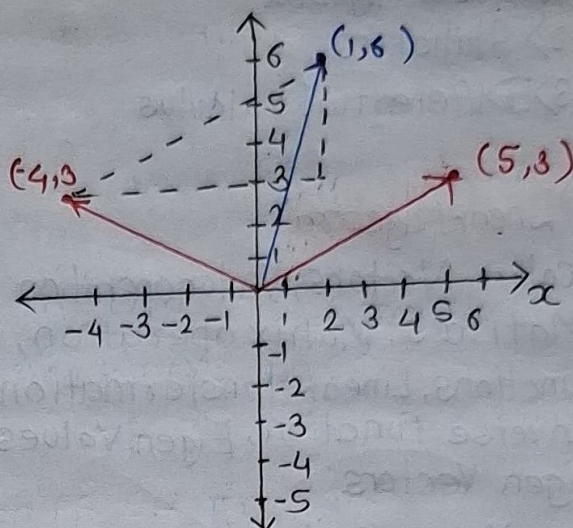


$\hat{i}, \hat{j} \rightarrow$ unit vector towards x and y axis $\Rightarrow 1$

* Addition of 2 Vectors

$$P_1 = \begin{bmatrix} -4 \\ 3 \end{bmatrix} \quad P_2 = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$P_1 + P_2 = \begin{bmatrix} -4 \\ 3 \end{bmatrix} + \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$$



Examples

:- Data Aggregation Task
 :- Feature Engineering

* NLP

Reviews	Sentiment
The product is good	1
The product is bad	0
...	...

word embedding :-

ex. Data: $[0.2, 0.1, 0.4]$

Science $[0.3, 0.7, 0.2]$

Data + Science = $[0.5, 0.8, 0.6]$

Image processing

RGB to GrayScale.

* Multiplication of Vectors

- 1) Dot product
- 2) Elementwise multiplication
- 3) Scalar multiplication.

Dot product

It is dot product of 2 vectors results in a scalar and is calculated as the sum of the products of their corresponding components.

$$A = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad B = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$A \cdot B = 2 \times 4 + 3 \times 5 \\ = 8 + 15 = 23$$

$$A \cdot B^T = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 4 & 5 \end{bmatrix} = 2 \times 4 + 3 \times 5 = 23$$

$$\vec{A} \cdot \vec{B} = (\text{length of projection } \vec{B}) \cdot (\text{length of vector } \vec{A})$$

$$\vec{A} \cdot \vec{B} = 0 \quad \text{Project the vector in the origin.}$$

* Application of Dot product in data science.

1] Cosine Similarity (RAG)

It is measure used to determine how similar 2 vectors are. It calculate the cosine of angle between two Vectors, providing a similarity score that ranges -1 (dissimilar) to 1 (complete Similar)

$$\cos \theta = \frac{A \cdot B}{|A| |B|} \quad \text{Dot product of } A \cdot B$$

* Recommendation System.

Action $\rightarrow$ Recommendation of other Action movie.

ex. Avenger: $[1, 2, 0, 3, 1]$

comedy $\rightarrow$ thrill

Action $\rightarrow$ Drama $\rightarrow$ Romance

$$B = [2, 0, 1, 1, 1]$$

Step 1:- Dot product ($A \cdot B$)

$$A \cdot B = 1 \times 2 + 2 \times 0 + 0 \times 1 + 3 \times 1 + 1 \times 1 = 6$$

Step 2:- $|A| |B|$

$$|A| = \sqrt{1^2 + 2^2 + 0^2 + 3^2 + 1^2} = \sqrt{15} \\ = 3.872$$

$$|B| = \sqrt{2^2 + 0^2 + 1^2 + 1^2 + 1^2} = \sqrt{7} \\ \approx 2.646$$

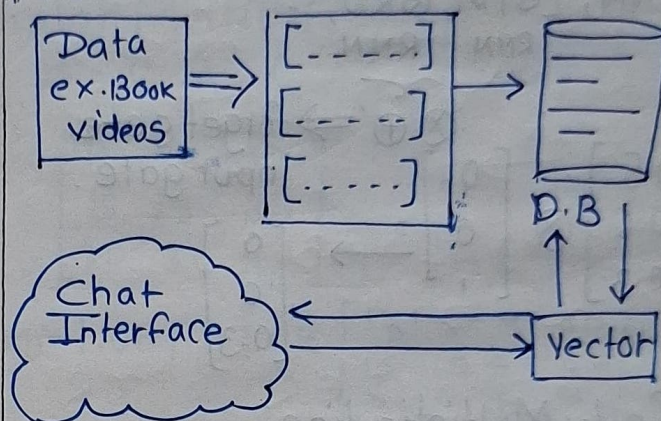
$$|A| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ \rightarrow \text{Euclidean Distance}$$

$$\cos \theta = \frac{6}{3.872 \times 2.646} = 0.586$$

$\therefore 58.6 \rightarrow$ Similar to avengers.
+ve similarity

movie 'B' will be recommended.

* Vector Data base



Cosine Similarity Search $\begin{bmatrix} \text{---} \end{bmatrix} \Leftrightarrow \begin{bmatrix} \text{---} \end{bmatrix}$
'A' $\leftrightarrow$ 'B'
 $\rightarrow 90\%$ similarity

2] Elementwise Multiplication

In elementwise multiplication, corresponding element of 2 vectors are multiplied to form a new vector of the same dimension.

$$A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad B = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$$

$$A \otimes B = \begin{bmatrix} 3 \\ 8 \\ 15 \end{bmatrix}$$

• Feature Engineering

Product	Cost	Discount
A	1000	0.1
B	500	0.2
C	200	0.15

⇒ two features

A) Discounted price	B) Final price
$\begin{bmatrix} 100 \\ 100 \\ 30 \end{bmatrix}$	$\begin{bmatrix} 900 \\ 400 \\ 170 \end{bmatrix}$

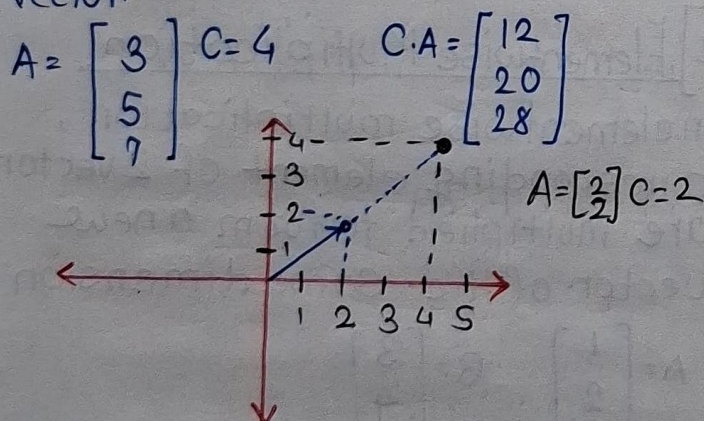
• Deep Learning

• RNN, LSTM, GRU, RNN RNN

$$\begin{bmatrix} 0.5 \\ 0.6 \\ 0.3 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 \\ 0 \\ 0.3 \end{bmatrix} \Rightarrow \text{forget gate, input gate.}$$

3] Scalar Multiplication

It involves multiplying vector by a scalar, resulting in a vector where each component is scaled by the vector



ex. Normalization and Standardization.

* Matrix

Matrix is a rectangular array of numbers, symbols or expression assigned in rows and columns.

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n}$$

has a_{ij} where i denotes the row and j denotes the columns.

Examples of matrices in Data Science

① Data Representation.

Math score	Phy score	Bio score
[55	65	75]
[65	60	55]
[70	45	80]

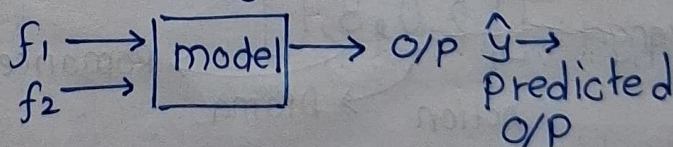
⇒ $\begin{bmatrix} f_1 & f_2 & f_3 \\ 55 & 65 & 75 \\ 65 & 60 & 55 \\ 70 & 45 & 80 \end{bmatrix}_{3 \times 3}$

② Images in Computer vision

Image =	$\begin{bmatrix} 0 & 128 & 255 \\ 255 & 128 & 0 \\ 128 & 255 & 128 \end{bmatrix}$	→ $\begin{bmatrix} 0 & 128 & 255 \\ 255 & 128 & 0 \\ 128 & 255 & 128 \end{bmatrix}$
(Gray scale)		

③ Confusion Matrix

∴ Accuracy of model



$y \rightarrow \text{actual output.}$

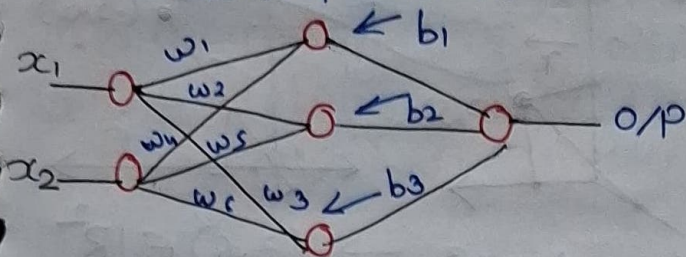
Confusion Matrix = $\begin{bmatrix} 50 & 10 \\ 5 & 35 \end{bmatrix}_{2 \times 2}$

50 → T.P 5 → F.P
10 → F.N 35 → T.N

$$\frac{TP + TN}{TP + FN + FP + TN} = \text{Accuracy}$$

④ Neural Network

:- Matrix operation.



No. of Study Hrs (x_1)	IQ (x_2)	Score (O/P feature)
4	100	90
5	90	85

$$Z = W^T x + b$$

$$W = \begin{bmatrix} w_1 & w_2 & w_3 \\ w_4 & w_5 & w_6 \end{bmatrix} \cdot x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} + \begin{bmatrix} b_1 & b_2 & b_3 \end{bmatrix}$$

• Regression.

$$y = mx + c$$

$$y = m_1 x_1 + m_2 x_2 + \dots \rightarrow m^T x + c$$

$$m = [m_1, m_2]$$

$$x = [x_1, x_2]$$

⑤ NLP (Natural Language Processing)

Sentiment analysis.

Review: The food is bad. Positive/Negative: 0

The food is good. Positive/Negative: 1

$$\begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.4 & 0.5 & 0.6 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

* Matrix Operation.

- 1) Matrix Addition and Subtraction
- 2) Scalar Matrix multiplication.
- 3) Matrix Multiplication.

Matrix addition and subtraction.
:- (2 matrices of same dimension)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad B = \begin{bmatrix} 4 & 5 & 6 \\ 7 & 8 & 9 \\ 1 & 2 & 3 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 5 & 7 & 9 \\ 11 & 13 & 15 \\ 8 & 10 & 12 \end{bmatrix}$$

Scalar Multiplication:-

$$B = C A \quad (\because C \text{ is scalar})$$

ex:-

$$P = \begin{bmatrix} 10 & 20 & 30 \\ 15 & 25 & 35 \\ 20 & 30 & 40 \end{bmatrix}$$

$$P_{adj} = 1.05 P = 1.05 \begin{bmatrix} 10 & 20 & 30 \\ 15 & 25 & 35 \\ 20 & 30 & 40 \end{bmatrix} = \begin{bmatrix} 10.5 & 21 & 31.5 \\ 15.75 & 26.25 & 36.75 \\ 21 & 31.5 & 42 \end{bmatrix}$$

Matrix Multiplication

:- It involves dot product of rows of first matrix with columns of second matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}_{1 \times 3} \quad B = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}_{3 \times 1}$$

$$\therefore 1 \times 2 + 2 \times 3 + 3 \times 4 = 20$$

ex.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2 \times 3} \quad B = \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}_{3 \times 2}$$

$$C_{11} = (1 \times 7) + (2 \times 9) + (3 \times 11) = 7 + 18 + 33 = 58$$

$$C_{12} = (1 \times 8) + (2 \times 10) + (3 \times 12) = 64$$

$$C_{21} = (4 \times 7) + (5 \times 9) + (6 \times 11) = 139$$

$$C_{22} = (4 \times 8) + (5 \times 10) + (6 \times 12) = 154$$

* Functions and Linear Transformation.

function:- A function is mathematical relationship that uniquely associate element of one set (called the domain) with element of another set (codomain) maps inputs to outputs.

Notation:- A function of mapping elements from set X (domain) to set Y (codomain)

$$f: X \rightarrow Y$$

ex. $f(x) = 2x + 3$

$$f: X \rightarrow Y$$

$f(x)$ mapping $2 \in \mathbb{R}$ to $7 \in \mathbb{R}$
 $f(2) = 2 \times 2 + 3 = 7$

Dimensionality Reduction.

$$f: \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \rightarrow \begin{bmatrix} x+y \\ 6z \end{bmatrix} \in \mathbb{R}^2$$



3 dimension $\rightarrow$ 2 dimension.

$$f: \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

Domain

Codomain.

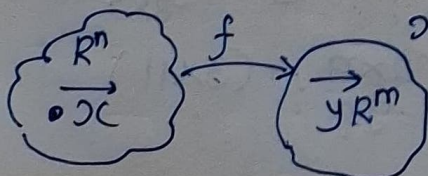
* Vector Transformations

$$f: X \rightarrow Y$$

$$\vec{x} \rightarrow \vec{y}$$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$$

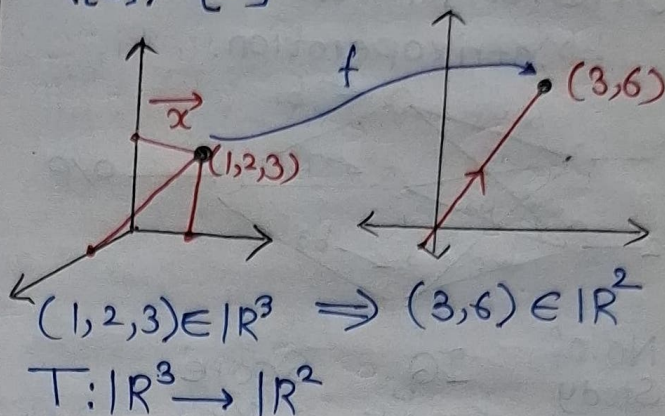
$$x_1, x_2, x_3, \dots, x_n \in \mathbb{R}$$



$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

ex. $f(x, y, z) = (x+y, 2z)$

$$f = \left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right) = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$



Vector transformation

vector transformation refers to operation that map vectors from one space to another, often changing their magnitude, direction or both.

These transformation are typically described using matrices and are fundamental in various fields including physics, engineering.

ex.

1) Scaling
 $v' = 2v$

2) Data normalization

3) Computer graphics.

4) Rotation:- Transformation that turns vector around origin.

eg:- rotation will be used in image processing

5) Reflection:- transformation that flips vector over a specified axis or plane.

① Linear Transformation.

a L.T is function between two vector spaces that preserves the operations of vector addition and scalar multiplication

This means that 'T' is a L.T from a vector space 'V' to vector space 'W'

It must hold

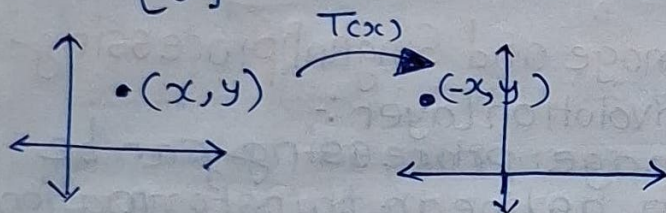
i) Additivity: $T(u+v) = T(u) + T(v)$

ii) Homogeneity: $T(cu) = cT(u)$

$u, v \in$ vector & c is a scalar value.

Eg. Reflection

$$x = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \quad T(x) = \begin{bmatrix} -x \\ y \end{bmatrix}$$



$$T(x) = AX = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -x \\ y \end{bmatrix}$$

① checking Additivity

$$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \quad v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$T(u+v) = T(u) + T(v)$$

$$u+v = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} u_1+v_1 \\ u_2+v_2 \end{bmatrix}$$

$$T(u+v) = A(u+v) = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1+v_1 \\ u_2+v_2 \end{bmatrix} = \begin{bmatrix} -(u_1+v_1) \\ u_2+v_2 \end{bmatrix}$$

$$T(u) = AU = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} -u_1 \\ u_2 \end{bmatrix}$$

$$T(v) = AV = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -v_1 \\ v_2 \end{bmatrix}$$

$$R.H.S = T(u) + T(v) = \begin{bmatrix} -u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} -v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -(u_1+v_1) \\ u_2+v_2 \end{bmatrix}$$

$$L.H.S = R.H.S$$

② checking Homogeneity

$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \in \mathbb{R}^2$ and c is scalar.

$$T(cu) = cT(u)$$

$$cu = c \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} cu_1 \\ cu_2 \end{bmatrix}$$

$$T(cu) = A(cu) = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} cu_1 \\ cu_2 \end{bmatrix} = \begin{bmatrix} -cu_1 \\ cu_2 \end{bmatrix} \rightarrow L.H.S$$

$$cT(u) = c(A \cdot u) = c \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = c \begin{bmatrix} -u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} -cu_1 \\ cu_2 \end{bmatrix} \rightarrow R.H.S$$

$$L.H.S = R.H.S$$

Examples that don't follow Linear transformation.

$$b = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad T(x) = x + b \quad T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x) = x + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

① Additivity

$$T(u+v) = T(u) + T(v)$$

$$u = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad v = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

$$T(u+v) = T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ -1 \end{bmatrix}\right) = T\left(\begin{bmatrix} 6 \\ 2 \end{bmatrix}\right)$$

$$T\left(\begin{bmatrix} 6 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 6 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \end{bmatrix} \rightarrow L.H.S$$

$$T(u) = \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$T(v) = \begin{bmatrix} 4 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

$$T(u) + T(v) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} + \begin{bmatrix} 5 \\ 0 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix} \rightarrow R.H.S$$

$$L.H.S \neq R.H.S$$

Homogeneity

$$T(Cu) = C T(u)$$

$$u = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad C = 2$$

$$T(u) = T \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ 6 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix} \rightarrow \text{L.H.S}$$

$$C T(u) = 2 \left(\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) = 2 \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \end{bmatrix} \rightarrow \text{R.H.S}$$

$$T(x) = \lambda + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow \text{Not a Linear Transformation.}$$

Why Linear Transformation?

1. Dimensionality Reduction.

(PCA)

It involves finding set of orthogonal axes and projecting data into axes. The transformation of data points in the original space to the new space defined by Principle component is a linear transformation.

- Reducing computational cost
- Mitigating the curse of dimensionality
- Visualizing high-dimensional data

2. Feature Engineering

It is used to create new feature from existing one.

Linear Regression, Ridge regression, linear discriminant analysis (LDA) all rely on L.T.

3. Data preprocessing

Normalization and standardization.

Standardization transforms data to have a mean of zero and a standard deviation of one while, normalization scales data to a specific range. These transformations are essential for ensuring that all features contribute equally to the model e.g. Gradient Descent.

3

4. Neural Networks.

In linear neural networks, especially deep learning models, the layers consist of linear transformation that map input data to intermediate layers, and eventually to the output layer.

5. Image and Signal processing

Convolution Layer:

In image processing can be seen as linear transformation applied to local regions of an image, Fourier transforms, which decompose signals into sinusoidal components are linear transformation that convert time-domain signals into frequency.

6. Optimization and solving system of linear equation.

Matrix inversions and the use of pseudo-inverse are essential for finding solution in linear regression.

7. Theoretical foundation.

8. Robustness and stability

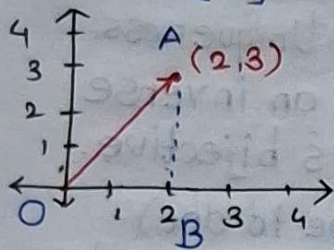
9. Geometric Interpretation

* Magnitude and unit Vector.

unit-vector: a vector has length of 1 mag.

Vector length :-

$$A = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \in \mathbb{R}^2$$



$$OB = |\vec{OA}|$$

$$= \sqrt{(OA)^2 + (OB)^2}$$

$$= \sqrt{x_1^2 + y_1^2}$$

$$|\vec{OA}| = \sqrt{4 + 9} = \sqrt{13}$$

Unit Vector:-

$$|\vec{u}| = 1 (\hat{u})$$

$$\vec{u} = \frac{1}{|\vec{v}|} \cdot \vec{v}$$

Scalar multiplication.

forex. $\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \in \mathbb{R}^3$

$$|\vec{v}| = \sqrt{1^2 + 2^2 + 0^2} = \sqrt{5}$$

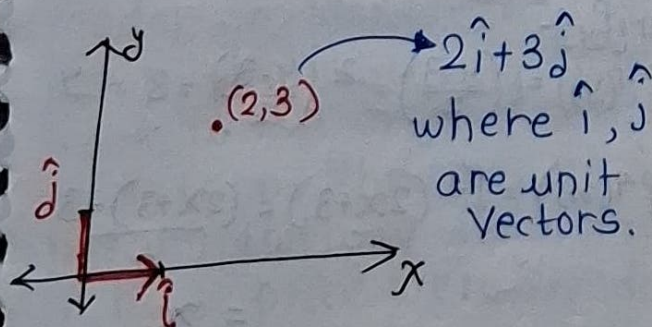
$$\vec{u} = \frac{1}{\sqrt{5}} \cdot \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \\ 0 \end{bmatrix}$$

$$|\vec{u}| = \sqrt{\left(\frac{1}{\sqrt{5}}\right)^2 + \left(\frac{2}{\sqrt{5}}\right)^2 + 0^2}$$

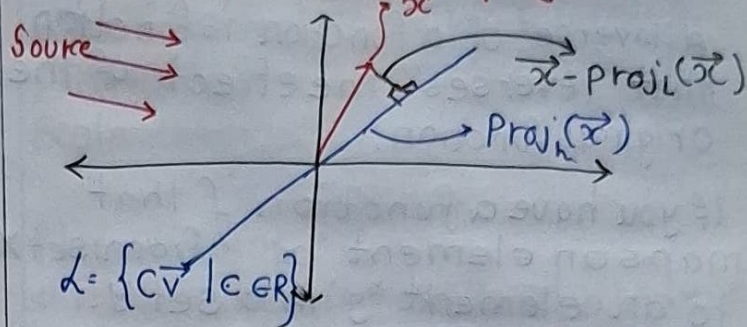
$$= \sqrt{\frac{1}{5} + \frac{4}{5} + 0}$$

$$= \sqrt{\frac{5}{5}} = 1$$

Hence, proved.



* Introduction to projection.

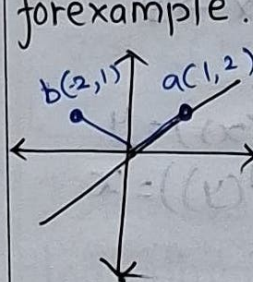


$\text{Proj}_L(\vec{x}) \rightarrow$ projection of $\vec{x}$ on the line L

pro

$\rightarrow$ Some vector in line where $\vec{x} - \text{Proj}_L(\vec{x})$ is $\perp$ to L

forexample.



$$a = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$a \cdot b = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$a \cdot b = (1)(2) + (2)(1)$$

$$a \cdot b = 0$$

Dot product is always Zero.

$\text{Proj}_L(\vec{x}) \rightarrow$

$$(\vec{x} - c\vec{v}) \cdot \vec{v} = 0$$

$$\vec{x} \cdot \vec{v} - c\vec{v} \cdot \vec{v} = 0$$

$$\vec{x} \cdot \vec{v} = c\vec{v} \cdot \vec{v}$$

$$c = \frac{\vec{x} \cdot \vec{v}}{\vec{v} \cdot \vec{v}}$$

$$\text{Proj}_L(\vec{x}) = c\vec{v} = \left(\frac{\vec{x} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \right) \vec{v}$$

forex.

$$L = \left\{ c \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} \mid c \in \mathbb{R} \right\} \quad \vec{x} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\text{Proj}_L(\vec{x}) = \left(\frac{\begin{bmatrix} 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix}}{\begin{bmatrix} 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix}} \right) \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2.8 \\ 1.4 \end{bmatrix}$$

* Inverse of a function.

A Inverse of a function is function that 'reverses' the effect of the original function.

If you have a function f that maps an element ' x ' from set X to an element ' y ' in a set Y . the inverse function f^{-1} map ' y ' back to ' x '.

Defination:-

Given a function $f: X \rightarrow Y$, then inverse function $f^{-1}: Y \rightarrow X$.

The inverse function $f: X \rightarrow Y$ has to satisfies

- 1) for all $x \in X: f(f^{-1}(x)) = x$
- 2) for all $y \in Y: f^{-1}(f(y)) = y$

* Identity function.

$$I_X: X \rightarrow X \quad I_X(a) = a$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$I.v = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

For set X , the identity function.

I_X is defined as

$$I_X(a) = a \quad \text{for all } a \in X$$

* Properties of Identity Function.

1) Preservation:- does not alter any element, If x is the domain then the image of x under the identity function is ' x '.

2) Linearity:- Identity function is linear transformation

$$I(u+v) = I(u) + I(v)$$

$$I(cu) = cI(u) = cu.$$

* Identity Matrix

$n \times n$

all diagonal element will be '1' and others are '0'

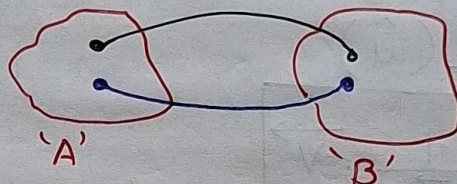
$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

* Existence and Uniqueness.

A function has an inverse if and only if it's bijective.

1) Injective (one to one)

:- different element in the domain is map to different element in the codomain.



2) Surjective (onto)

:- Every element in the Codomain is the image of at least one element in the domain.

for ex :-

$$f(x) = 2x+3$$

$$y = 2x+3 \quad \text{for } x:$$

$$y-3 = 2x$$

$$x = \frac{y-3}{2}$$

The inverse of function is

$$f^{-1}(y) = \frac{y-3}{2}$$

Verify:-

$$f(f^{-1}(y)) = f\left(\frac{y-3}{2}\right) = 2\left(\frac{y-3}{2}\right) + 3 = y$$

$$f^{-1}(f(x)) = f^{-1}(2x+3) = \frac{(2x+3)-3}{2} = x$$

* Applications of Inverse Function.

1] Normalization and Standardization
for example:-

Area	No. of rooms	Dependent feature ↓	
		Area (sq.ft)	Price
1500	4	1800	50L
1800	3	2100	60L

Independent feature.

We have to scale down the feature.

Standardization :- $\mu = 0$ $\sigma = 1$

Standard Normal Distribution.

No. of Rooms $\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \in \mathbb{R} \xrightarrow{\text{Transformation}} Z = \frac{x_i - \mu}{\sigma}$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \xrightarrow{f(x)} \begin{bmatrix} -1.5/2 \\ -0.5/2 \\ 0.5/2 \\ 1.5/2 \end{bmatrix} \xrightarrow{f^{-1}(x)} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$Z_i = \frac{x_i - \mu}{\sigma}$

$\mu = \frac{1+2+3+4}{4} = 2.5$ $\sigma = 2$

Original transformation

$$Z = \frac{x - \mu}{\sigma}$$

Inverse transformation.

$$x = Z\sigma + \mu$$

Use Case:-

After training Machine learning model on standardized data. the prediction are often rescaled back to the original scale to interpret the result in a meaningful way. for instance if house price were standardized.

the inverse transformation would convert the standardized prediction back to original price scale

Normalization.

:- Feature scaling with Min Max Normalization.

Original transformation.

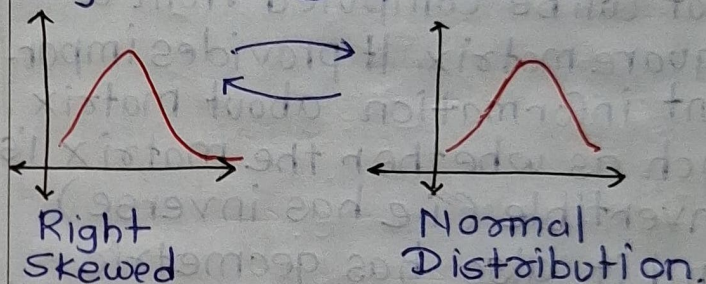
$$Z = \frac{x - \min(x)}{\max(x) - \min(x)} \quad T: x \rightarrow y$$

Inverse transformation:-

$$x = \frac{Z(\max(x) - \min(x))}{x - \min(x)} \quad T^{-1}: y \rightarrow x$$

2] Distribution of Data.

logarithmic Distribution.



original transformation :- $y = \log(x)$

Inverse transformation :- $x = e^y$

Use Case:-

In financial data analysis, income or sales data often exhibit skewness. Applying a log transformation can stabilize the variance and make pattern more visible. After model prediction, the inverse log transformation is applied to interpret result on the original scale.

Data Encryption and Decryption.

Use Case :-

Decryption is applied to retrieve original information.

* How to find Inverse of a Matrix

① Determinant

② Inverse.

for ex:- 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\det(A) = ad - bc$$

The determinant is scalar value that can be computed from a square matrix. It provides important information about matrix such as whether the matrix is invertible (i.e. has inverse) and it also has geometric interpretations, such as describing scaling factor of linear transformation represented by matrix.

transformation using A
 $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \rightarrow x$ using A then A^{-1} to recover the original vector.

$$y = AX = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4+7 \\ 2+6 \end{bmatrix} = \begin{bmatrix} 11 \\ 8 \end{bmatrix}$$

Recovering x using A^{-1}

$$x = A^{-1}y = \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix} \begin{bmatrix} 11 \\ 8 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Thus A^{-1} successfully recover the original vector x .

* Eigen Values and Eigen Vectors.

Eigen values and eigen vectors are fundamental concepts of Linear Algebra.

They provide insights into the properties of linear transformation, represented by matrices

Eigen Value (λ)

A scalar that indicates how much an eigen vector is stretched or compressed during Linear Transformation.

Eigen Vector (v)

A non zero vector that only change in scale (Not direction) when a linear transformation is applied.

$$AV = \lambda V$$

for sq. matrix ' A ' an eigen vector and its corresponding eigen value λ satisfy the above eqn.

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

1) Find eigen Values

$$\det(A - \lambda I) = 0$$

$$\begin{aligned} A - \lambda I &= \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \\ &= \begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix} \end{aligned}$$

$$\begin{aligned}\det(A - \lambda I) &= (4 - \lambda)(3 - \lambda)(2)(1) \\ &= (4 - \lambda)(3 - \lambda) - 2 \\ &= 12 - 4\lambda - 3\lambda + \lambda^2 - 2 \\ &= \lambda^2 - 7\lambda + 10\end{aligned}$$

Solve quadratic eqⁿ

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a=1 \quad b=-7 \quad c=10$$

$$\lambda = \frac{7 \pm \sqrt{49 - 40}}{2}$$

$$= \frac{7 \pm \sqrt{9}}{2}$$

$$\lambda_1 = \frac{10}{2} \quad \lambda_2 = \frac{4}{2}$$

$$\lambda_1 = 5 \quad \lambda_2 = 2$$

Eigen values of A is $\lambda_1 = 5$
and $\lambda_2 = 2$

Find eigen Vector

$$(A - \lambda I)V = 0$$

for $\lambda_1 = 5$

$$A - 5I = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$-x + y = 0$$

$$\boxed{y = x}$$

for $\lambda_2 = 2$

$$A - 2I = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$2x + y = 0$$

$$\boxed{y = -2x}$$

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} \quad \lambda_1 = 5 \quad \lambda_2 = 2$$

$$\text{for } \lambda_1 = 5 \quad V_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

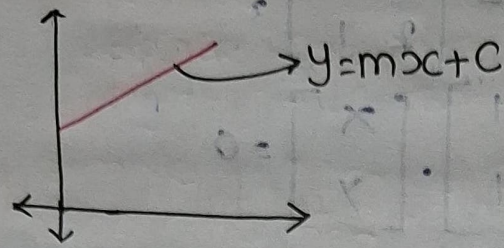
$$\text{for } \lambda_2 = 2 \quad V_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

These eigen vectors and eigen values describe how the matrix 'A'. 'A' scales and rotate Vectors in its transformation. Eigen Values indicate the factor by which the eigenvectors are stretched or compressed and eigen vectors provide directions in which this stretching or Compression occurs.

*Application of Eigen Vector and Eigen Values

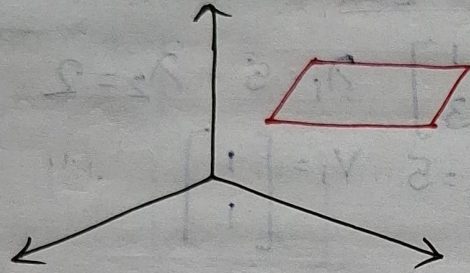
- Principle Component Analysis (PCA)
- Dimensionality Reduction.

* Equation of Line, 3D plane and Hyperplane.



$$w_1 x_1 + w_2 x_2 + b = 0$$

$w^T x + b = 0$ - equation of Line.



$$w_1 x_1 + w_2 x_2 + w_3 x_3 + b = 0$$

$$w^T x + b = 0$$

$$w = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

n-Dimensional plane.

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_n x_n + b = 0$$

$$w^T x + b = 0$$

Equation of a straight line passing through origin

$$w^T x = 0$$

Equation of plane.

$$w^T x = 0$$

$$w \cdot x = w^T x = \|w\| \|x\| \cos \theta = 0$$

$$\theta = 90^\circ$$

$$\cos 90^\circ = 0$$